

Supplementing Patient Encounter Training for Preservice Nurses: Towards an AI×VR Approach

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Abstract—We built an immersive virtual reality environment where preservice nurses practice with generative artificial intelligence-driven virtual standardized patients (AI-VSPs). We ran a mixed-methods study (experimental $n = 23$; control $n = 34$). Usability score was 81.6/100, indicating excellent usability. A single session increased self-efficacy from 3.74 to 4.11 ($t(22) = 3.82$, $p < 0.001$). In the standardized-patient (SP) encounter performance assessment, scores for AI-VSP users did not differ from controls ($t(55) = -0.134$, $p = 0.894$). AI-VSP encounter performance did not relate to engagement, presence, or immersion, although these experience measures were strongly intercorrelated. Interviews highlighted helps and hindrances. Overall, AI-VSPs are a usable, confidence-building supplement to SP encounter preparation. We also outline practical design steps: cap reply length, enable reliable barge-in, improve speech quality and latency, and keep scaffolding minimal.

Keywords—artificial intelligence, virtual reality, nursing education, standardized patient.

I. INTRODUCTION

The World Health Organization projects a global shortfall of 4.5 million nurses by 2030 [1]. In the United States, demand continues to rise even as schools struggle to expand enrollment due to faculty shortages, limited clinical placements, and constrained facilities [2], [3]. As a result, and given its pedagogical value, simulation-based education has moved to the center of many curricula, offering safe, repeatable practice across domains from mental health to end-of-life care and interprofessional communication [4]. This approach is consistent with long-standing guidance to recreate real-world scenarios for interactive learning and feedback [5], [6].

A cornerstone of communication training is the standardized patient (SP): trained actors who portray cases for deliberate practice and objective structured clinical examination [7], [8]. SPs provide realism and formative feedback but are resource-intensive to schedule and oversee, and are susceptible to variability and fatigue, which can limit scalability and consistency [8], [9]. These constraints motivate scalable supplements that preserve authentic conversation while increasing access.

Two technological shifts renew this opportunity. First, immersive Virtual Reality (VR) provides embodied presence and spatial interaction while standardizing exposure [10]; reviews report gains in knowledge, skills, and engagement

across health [10] and higher education and highlight design opportunities for embodied interaction [11], [12]. Second, generative artificial intelligence (AI) enables open-ended, persona-consistent conversation that better approximates human dialog than rule-based pipelines; early deployments in health education and dentistry report promising usability and perceived realism, while also noting the need to manage accuracy, verbosity (over-long replies), and latency [13], [14], [15]. Together, VR and AI suggest a path to AI-driven virtual standardized patients (AI-VSPs) that combine authentic, unscripted interaction with the repeatability and scalability of simulation.

Building on this rationale, we asked whether an AI-VSP-based learning system can deliver usable, authentic practice while addressing the scalability limits of traditional SP encounters. This work presents an immersive VR environment that enables preservice nurses to practice patient encounter skills with AI-VSPs in a single-user clinic with speech-based, open-ended conversations, varied patient personas, and structured assistance (captions and cue sheets) to support skills coverage without over-structuring. We quantified usability, compared SP encounter performance between experimental and control groups, assessed within-learner change in self-efficacy from pre to post intervention, and tested associations between AI-VSP encounter performance and post-session engagement, presence, and immersion. Semi-structured interviews further identified agent- and system-level factors that helped or hindered learning and confidence. Accordingly, we posed the following research questions (RQs):

RQ1: How do preservice nurses rate the usability of the AI-VSP?

RQ2: How do SP encounter performance compare between students who practice with AI-VSPs and those who use traditional preparation?

RQ3: How does self-efficacy change from pre- to post-AI-VSP session?

RQ4: How is AI-VSP encounter performance associated with post-session engagement, presence, and immersion?

RQ5: How do students describe agent and system features (e.g., agent behaviors, persona fit, in-scene scaffolds, and usability factors) as helping or hindering learning, confidence, and perceived transfer?

II. RELATED WORK

A. Immersive VR for Experiential Learning and Clinical Communication

Immersive VR is commonly defined as a head-mounted display (HMD), motion-tracked system that renders interactive 3D scenes and maps user movements to real-time audiovisual feedback [16], [17]. Classic framings locate VR at the intersection of interaction, immersion, and imagination [18]. Across domains, VR has supported collaborative gameplay and problem solving [19], high-risk skills training (e.g., firefighting and defense) [20], [21], health/anatomy practice [22], geology knowledge understanding [23], and engineering procedures and laboratories [24], [25]. Reported advantages include safe, repeatable experiential practice and heightened motivation and engagement that can stimulate exploration and creativity [26], [27], [28].

Learning outcomes in VR are moderated by cognitive load and agency. Well-scaffolded experiences can increase affect and engagement, whereas excess sensory or task demands inflate mental workload and reduce performance [27], [29]. Presence can deepen involvement but, when unmanaged, may add load; designs that preserve user control and self-pacing tend to fare better [30], [31]. In nursing and allied health, immersive and multiuser environments increase participation and perceived value when tasks are authentic and interaction is meaningful [32]. Systematic reviews characterize immersive VR in higher education as effective and scalable when key design elements (feedback, authenticity, pacing) are explicit [33]; meta-analyses in nursing echo gains in knowledge and skills with strong learner satisfaction [34], [35], while integrative and qualitative reviews catalog virtual-specific design characteristics for engagement and perceived authenticity [36], [37].

B. From Standardized Patients to Virtual Standardized Patients

Patient-encounter training builds communication, information-gathering, and clinical reasoning skills foundational to safe practice. SP encounters are widely used across medical and nursing programs for formative and summative evaluation, but they are resource-intensive and logistically demanding, which limit scalability and raise feasibility and reliability/validity concerns [7], [38], [39], [40]. In addition, first-time SP users often experience nerves and difficulty “suspending reality,” highlighting variability in learner experience and added program burden [38]. These constraints motivate virtual standardized patients (VSPs)—avatar-based agents that enable on-demand, repeatable interviewing aligned with experiential learning principles [41], [42], [43].

VSPs have supported history-taking for novice and advanced learners [44], [45], [46]. Early generations, however, frequently relied on scripted menus or keyword triggers, limiting coverage and adaptability to unanticipated queries; reduced conversational fidelity and uneven usability, in turn, constrained authenticity and transfer to assessment settings [47], [48], [49]. Immersive VR has also been used to deliver standardized mental-health encounters at scale, improving access and consistency, but common constraints persist—notably menu-driven dialogue, limited nonverbal expressivity, uncanny text-

to-speech (TTS) voice and prosody (speech rhythm and intonation), and interaction-mechanic workload [50].

C. AI for Conversational Patients and AI×VR Integration

AI is playing an increasingly prominent role in education, offering new ways to personalize instruction, assess learning, and simulate real-world interactions. In nursing and healthcare education, AI shows promise in enhancing clinical reasoning, communication skills, and learner engagement through generative tools such as large language models (LLM) [51]. In higher education more broadly, AI has been applied to tutoring, feedback, and adaptive systems, though questions remain about its impact on educators’ roles and instructional design [52]. These opportunities are accompanied by ethical challenges. Studies highlight the importance of ensuring transparency in AI systems, mitigating bias in model outputs, and protecting learner data [53]. As these tools evolve, scholars call for policy frameworks and governance strategies to support responsible implementation [54]. These considerations are especially relevant when integrating AI into VR, where conversational AI agents can be used to simulate realistic patient interactions in safe, repeatable training scenarios.

In nursing, a text-chat LLM patient yielded high engagement and perceived value [55], and work outside nursing shows AI can improve performance and user experience in programming and game-based learning [56], [57]. In anatomy education, an embodied AI assistant in VR improved knowledge-level performance relative to higher-order tasks, underscoring how cognitive complexity and embodiment shape outcomes [58].

Bringing AI into VR with speech requires coupling automatic speech recognition (ASR) and TTS with dialog models and embodied avatars. This integration promises more natural, adaptive interviews and richer case diversity, but also introduces design tensions—latency, verbosity/interruptibility, prosody, and the paucity of nonverbal signals—which shape workload, rapport, and empathy. Qualitative student perspectives on AI-VSP encounters report positive learning value alongside technical frictions and difficulty building empathy when voices sound synthetic or when control over turn-taking is limited, pointing to concrete interaction targets for next-generation systems [55].

Taken together, prior work motivates an AI×VR approach in which VR provides embodied, standardized practice environments and AI enables open-ended patient dialogue—supporting the promise of AI-VSPs for patient-encounter training. However, rigorous AI-VSP related studies that connect (i) usability and deliberate practice, (ii) immediate performance during AI-VSP encounters, (iii) pre/post self-efficacy, and (iv) transfer to physical SP encounter performance—while also eliciting trait-level explanations of which agent/system features help or hinder learning—remain limited. Our study addresses this gap.

III. MATERIALS AND METHODS

In the following sections, we describe the theoretical framework, apparatus, research design, instruments, participants, data collection, and data analysis.

A. Theoretical Framework

We grounded the study in transfer of training and deliberate practice to explain how AI-VSPs can promote generalization of patient-encounter and efficient improvement in performance.

Transfer concerns the extent to which skills acquired in instruction generalize to, and are retained in, the target environment. Classic accounts identify three determinants: learner characteristics such as prior skill and self-efficacy; training design features such as task fidelity, practice variability, and timely feedback with reflection; and the work setting, including cues and support that enable application. Near transfer is maximized when practice reproduces the critical cues and response demands of the criterion setting, whereas far transfer requires abstraction across dissimilar contexts [59], [60]. For patient-encounter training, designs that mirror real patient-encounter scenarios through history-taking cues and turn-taking constraints, provide variability across cases, and make feedback opportunities explicit are most likely to support transfer to SP encounter performance.

Deliberate practice specifies the microconditions for improvement: clear goals, tasks set at an appropriate level of challenge, immediate and informative feedback, repetition with focused refinement, and progressive difficulty as proficiency grows. Unlike unguided repetition, deliberate practice targets specific weaknesses and directs attention to task-relevant cues, producing reliable gains in complex skills, including clinical communication, over short cycles of practice and feedback [61], [62].

B. Apparatus

1) *Platform and hardware*: The immersive clinic was implemented in Unity3D [63] as a real-time 3D application and deployed to a Meta Quest 2 headset. Participants spoke through the headset microphone and used a push-to-talk (hold the trigger to speak) control mapped to the primary trigger. An on-screen indicator showed when the microphone was active.

2) *Conversational agent integration*: The virtual patient was created by integrating the Inworld API [64] with the clinic scene to couple ASR, LLM-based dialogue, and TTS. This integration proceeded in three stages—AI character creation, Unity import, and VR setup. AI character creation: for each virtual patient, we specified a TTS voice and avatar with a short scene description for context. Unity import: we authenticated the project to the Studio workspace, added the Inworld Controller, instantiated the character prefab in front of the player, mapped the primary trigger to push-to-talk, displayed live captions above the avatar for quick verification/repair, and spatialized TTS so the patient’s voice originated from the avatar’s position. VR setup: we targeted Meta Quest 2, used a lightweight Universal Render Pipeline profile in Unity with post-processing disabled for headset performance, referenced the XR camera in the Inworld Controller for correct

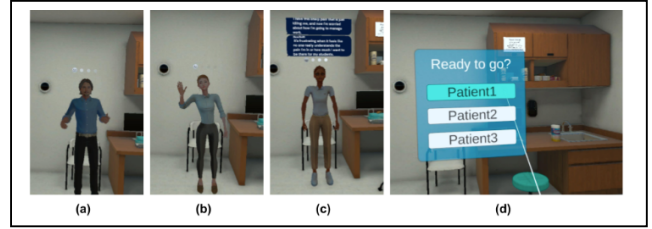


Fig. 1. (a,b,c) AI-VSPs with introverted, ambiverted, and extroverted personas; (d) floating patient-selection menu.*

head-related audio, and kept the interface minimal to preserve natural, open-ended speech interaction.

3) *Patient personas and case*: Three virtual patients were authored with distinct personalities—introvert, ambivert, and extrovert [65]—so learners practiced adapting to patients’ communication styles (Fig. 1). We worked closely with the professors at our university’s nursing school who have been teaching the related course for years, and finalized the patient profiles after multiple rounds of discussions with them. A floating menu allowed participants to select one patient each time (Fig. 1). Although appearances, voices, names, ages, and family details differed, each patient shared the same clinical profile of acute low-back pain radiating to the right ankle that began that morning, unrelieved by over-the-counter medication, and with no prior back pain history.

4) *Environment and interaction*: The clinic room contained common primary-care elements, including an exam table, a water sink, a desk with a computer, storage cabinets, and small supplies (Fig. 2). A minimal interface provided patient selection and session start/stop controls. No scores were shown during the interview.

5) *Transfer-informed scaffolds*: Two assistance mechanisms supported interviewing and near transfer: live captions rendered both learner and patient utterances above the avatar (Fig. 2) to aid verification and repair, and a floor cheatsheet (Fig. 2) listed core history-taking prompts—location, quality, duration, and aggravating or alleviating factors. Together, these cues foreground criterion-relevant features without over-scripting practice.

6) *Data capture and privacy*: The system is able to record each conversation start and end times, de-identified conversation logs, timestamps for each conversational turn, and patient selection. The conversations in the AI-VSP encounters can be converted to text by the speech engine instantly.



Fig. 2. Immersive VR clinic for AI-VSP encounters: (a) exam room with common primary-care elements; (b) learner view with floor “cheat sheet,” exit button (lower left), and a caption panel above the AI-VSP showing both the learner and patient utterances.*

*Similar, but distinct, screenshots may be found in other publications from the same research team.

C. Research Design

We used a mixed-methods, quasi-experimental design with a nonequivalent control group. In this collaborative study, professors from the School of Nursing at the same Midwestern R1 university as the authors designed the patient encounter training content to align with their nursing curriculum. This content included a learning scenario where nursing students gathered information from patients and used various communication techniques (e.g., self-introduction, questions about pain quality and location). Per the course instructional design, students completed a SP encounter as the patient-encounter assessment. AI-VSP practice was offered to all students as an optional preparatory activity prior to the SP encounter assessment. Students who opted in comprised the experimental group, whereas those who did not opt in formed a nonequivalent control group and prepared using the course's longstanding learning materials and activities.

We collected quantitative data with validated instruments aligned to our constructs and post-session experience. To probe adoption drivers and mechanisms behind these patterns, we conducted semi-structured individual interviews immediately after the AI-VSP encounter session and again post-SP encounter session, then applied content analysis. This convergent mixed-methods approach enabled triangulation and explanation of the survey/behavioral results by surfacing recurrent themes that shape the participant's insight of their experience with the AI-VSP encounter.

D. Instruments

To collect data, we adopted validated questionnaires, a structured performance rubric, and semi-structured interviews. Measures were administered at the time points described in the data collection section.

Usability (RQ1). Usability was assessed with the 10-item System Usability Scale (SUS) [66]. Items use a 5-point agreement scale and were converted to a 0–100 score using the standard rescoring procedure.

Performance (RQ2). Patient encounter performance was assessed with a 15-item checklist (Table I) developed and provided by the nursing school instructors. Each required element earned 1 point when elicited, and elements volunteered by the patient received credit if the learner accurately confirmed them. Two independent raters scored all encounters, and the primary score was the mean of their ratings. Inter-rater reliability was assessed using Cohen's Kappa [67].

Self-efficacy (RQ3). Learners completed the Motivated Strategies for Learning Questionnaire self-efficacy subscale [68] immediately before AI-VSP encounter and again after AI-VSP encounter to index readiness for the subsequent SP encounter.

Post-VR session experience (RQ4). Engagement, presence, and immersion were rated on 5-point Likert items. Presence constructs align with established presence factors [30]; incorporating facets of game-based learning immersion [69]; and engagement [70] with validated user-engagement dimensions.

TABLE I. RUBRIC FOR EVALUATING NURSING STUDENTS' PATIENT ENCOUNTER PERFORMANCE

Subdivision	Code	Short Description	Value
Clinical Assessment Items	C01	Explores patient's feelings	1
	C02	Identifies location of pain	1
	C03	Clarifies onset of pain	1
	C04	Assesses type/quality of pain	1
	C05	Assesses severity of pain	1
	C06	Reviews history of back problems	1
	C07	Identifies factors that relieve pain	1
	C08	Identifies factors that worsen pain	1
	C09	Assesses impact on function	1
	C10	Assesses emotional state	1
Greeting and Closure Items	G01	Provides appropriate greeting	1
	G02	Expresses courteous acknowledgment	1
	G03	Introduces self by name and role	1
	G04	Uses general well-being inquiry	1
	G05	Provides appropriate farewell	1

Qualitative interviews (RQ5). Semi-structured interviews probed perceptions of conversational quality, presence/immersion/engagement, usability, difficult personas, and perceived transfer. On-site interviews were conducted immediately after the AI-VSP encounter session; follow-up interviews occurred after the SP encounter assessment. Guides were standardized and used flexible probes as needed. Sessions were audio-recorded, transcribed, and de-identified.

1) Sample on-site questions (post-AI-VSP encounter):

- Is there anything you like or dislike regarding the VR experience?
- How much do you feel like you were in the real patient-encounter room during the VR experience?
- Did you feel like you had a sense of control over your learning during the VR experience?

2) Sample follow-up questions (post-SP encounter):

- In general, how would you compare your VR patient encounter experience and the physical one?
- Do you think the VR patient encounter experience affected your confidence in the physical one? Why or why not?
- Compared with the physical patient encounter, what activities or functionalities do you think are essential to be included in the VR experience but were actually missing?

E. Participants

The participants were nursing students from a midwestern R1 university in the US. Twenty-four students volunteered for the experimental group, and all 24 completed post-VR interviews. Complete survey data were obtained from 23

participants (21 females, 2 males). Most were between 20 and 23 years old (61%), with the remaining 39% aged 18–19.

An additional 39 students who did not take AI-VSP encounters were recruited as the control group. Complete data were obtained from 34, comprising 33 females and 1 male. Most participants in the control group were aged 20–23 years (76%), with the remainder aged 18–19 years (24%). The cohort size reflects a single class intake typical of simulation studies [8]. The research received approval from the university's Institutional Review Board, and the team obtained consent from the corresponding participants before data collection.

F. Data Collection

1) *Baseline measures and orientation (experimental group)*: AI-VSP encounters were hosted in university spatial classrooms to ensure single-user immersion (Fig. 3). Before the formal session, participants completed a pre-survey to report their demographic information and self-efficacy before the study; then, they received a brief headset orientation covering head fit, controls, and speaking to the AI-VSPs. Students practiced basic operations in the virtual environment until they felt comfortable, after which they proceeded to the AI-VSP encounters.

2) *Practice session (both experimental and control group)*: Each student in the experimental group completed a ~30-minute AI-VSP encounter session with three distinct patient personas; patient order was counterbalanced by participants' selection. The AI-VSP encounters were recorded using the Meta Quest headset's video and audio recording functions. Students in the control group prepared for the SP encounter assessment using long-standing course materials and activities.

3) *Immediate post-AI-VSP encounter measures (experimental group)*: Immediately after the AI-VSP encounter, experimental-group students completed a post-survey on the system usability, self-efficacy, engagement, presence, and immersion.

4) *Patient encounter assessment (both experimental and control group)*: Within about 48 hours of the AI-VSP encounter session window, both groups completed the SP encounter assessment (Fig. 3). Performances in the experimental group's AI-VSP encounter session and in both groups' SP encounters were scored with the same 15-item scoring rubric (Table I.) by the researchers. Two researchers were randomly assigned to score each encounter independently; the final score was the mean of both ratings, and inter-rater reliability was estimated with Cohen's Kappa.

5) *Qualitative study (experimental group)*: To contextualize the quantitative findings, we conducted semi-structured interviews after both the AI-VSP encounter session and the SP encounter performance assessment. All interviews were audio-recorded and transcribed using Zoom's transcript function, with manual review to ensure accuracy while maintaining efficiency.

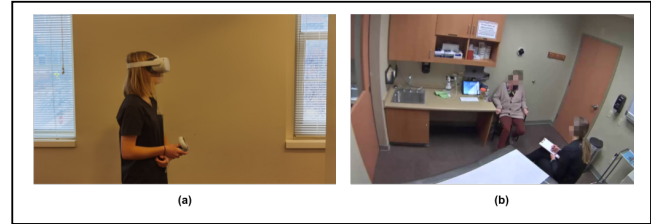


Fig. 3. (a) A preservice nurse practices patient encounters with AI-VSPs in a spatial classroom, enabling full individual immersion. (b) A preservice nurse completes a performance assessment with a SP at the nursing school.

G. Data Analysis

We analyzed quantitative and qualitative strands separately, then integrated them to explain observed effects.

Quantitative data: Descriptive statistics summarized SUS scores; SP encounter performance was compared between groups with an independent - samples t-test; pre - post self-efficacy change was tested with a paired-samples t-test; and Pearson correlations examined links between AI-VSP encounter performance and post-session engagement, presence, and immersion, as well as interrelations among the experience measures.

Qualitative data: Interviews were transcribed, then analyzed using Braun and Clarke's six-phase thematic analysis [71]. One researcher conducted the initial coding and theme development. To enhance validity, the full codebook and sample coded excerpts were shared with each of the research team members to review individually. Through collaborative discussions, the team refined the corresponding theme definitions and reached consensus on a final set of themes that reflected the breadth of participant perspectives.

IV. RESULTS

A. Usability (RQ1)

The SUS indicated acceptable usability for the learning platform in this cohort ($n = 23$). The SUS mean score of 81.6/100 ranks the usability of our created AI-VSPs noticeably above the third quartile, defining the tool as "Acceptable" and with an "Excellent" level of usability [66].

B. Performance (RQ2)

An independent sample t-test was conducted to compare the SP encounter performance scores between the experiment and control groups. Although there was a slight rise in the mean of performance in experiment group ($M = 11.28$, $SD = 1.06$) than control group ($M = 11.24$, $SD = 1.45$), there was no statistically significant difference in scores between the two groups, $t(55) = -0.134$, $p = 0.894$. Two independent raters scored all encounters, with acceptable inter-rater agreement (Cohen's Kappa = 0.70).

C. Self-Efficacy Change (RQ3)

We performed a dependent sample t-test on participants' self-efficacy scores before and after the AI-VSP encounter, revealing a significant increase in self-efficacy from $M = 3.74$ ($SD = 0.51$) to $M = 4.11$ ($SD = 0.57$), $t(22) = 3.82$, $p < 0.001$.

TABLE II. THE CORRELATION MATRIX OF AFFECTIVE CONSTRUCTS ($N = 23$)

Variables	1	2	3	4
Performance in VR	1.00			
Engagement	-0.17	1.00		
Presence	0.07	0.63**	1.00	
Immersion	-0.11	0.56**	0.83***	1.00

Note: ** Significant at $\alpha = 0.01$, *** Significant at $\alpha = 0.001$

D. AI-VSP Encounter Performance–Experience Associations (RQ4)

We examined correlations between AI-VSP encounter performance and post-session engagement, presence, and immersion. As shown in Table II, descriptively, engagement $M = 3.95$ ($SD = 0.41$), presence $M = 4.07$ ($SD = 0.82$), and immersion $M = 3.95$ ($SD = 0.56$). AI-VSP encounter performance did not correlate with engagement ($r = -0.17$), presence ($r = 0.07$), or immersion ($r = -0.11$). By contrast, the experience measures are intercorrelated: engagement–presence $r = 0.63$ ($p < 0.01$), engagement–immersion $r = 0.56$ ($p < 0.01$), and presence–immersion $r = 0.83$ ($p < 0.001$). These findings suggest that while participants generally felt moderately to highly engaged, present, and immersed while using the learning platform, and those feelings tended to reinforce one another, these subjective impressions did not correspond to how well they actually performed during the AI-VSP encounters. In other words, feeling immersed or engaged did not necessarily mean better performance in this sample.

E. Qualitative Mechanisms (RQ5)

The interviews in the study revealed three main themes: engaging learning, educational benefits of the AI-VSP encounters, and usability challenges.

Engaging learning. All participants (24) reported strong engagement and presence driven by the head-mounted clinic and unscripted dialog—e.g., “75% of the time I felt fully in the room” (P04); “felt very real” (P12); “I sometimes felt like I was actually in the room talking to a real patient” (P02).

Educational benefits. All students (24) described gains in redirecting digressions, eliciting sparse information, and sequencing questions, alongside higher confidence—e.g., “bring the conversation back to the main topic” (P03); “helped me build more confidence” (P06); “I had to assess and respond accordingly” (P07). These mechanisms align with the observed pre–post self-efficacy increase and perceived readiness for the SP encounter assessment.

Usability challenges. The majority of the participants (18/24) noted frictions that raised workload and sometimes limited empathy: latency/freezes (e.g., P06, P13), synthetic-sounding voices (P08), and overlong replies without easy interruption (P10, P23). Typical remarks included “It was unnecessary and irrelevant... it dragged on too long” (P02), “I couldn’t stop them; I wanted a clear stopping point” (P09), and “I didn’t like the voices” (P07). Within this theme, participants cited turn-taking problems and verbosity in 18/24 interviews; requested conversation controls in 14/24—pause/stop and barge-in (ability to interrupt the agent mid-utterance); and noted voice naturalness limits in 10/24 (synthetic quality and prosody, i.e., speech rhythm and intonation)—helping explain why

experience ratings (engagement, presence, immersion) were high yet did not correlate with AI-VSP encounter performance.

V. DISCUSSION

A. Usability of the AI-VSP

Addressing RQ1, SUS results indicated excellent usability (mean 81.6/100), suggesting the platform is operable enough to support routine practice while learners focus on task goals rather than interface management [66]. Students’ descriptions of straightforward controls and a focused, single-user clinic align with simulation design guidance that emphasizes authenticity, pacing, and minimal interface overhead to sustain operability and value [5], [33]. These strengths are consistent with meta-analytic reports of positive learner experience with immersive technologies in nursing and health education [34], [35]. From a transfer-of-training perspective, this level of usability is a prerequisite for near transfer: learners must be able to reproduce core interview cues and responses without excessive interface overhead [59], [60]. Deliberate practice likewise requires tight control of pacing and immediate, informative feedback; interaction frictions dilute these conditions by diverting attention from goal-directed refinement [61], [62].

However, qualitative feedback surfaced flow frictions—over-long replies, weak interruptibility, synthetic prosody, and occasional latency/freezes—that mirror known threats to conversational fidelity in VSPs [48], [49] and in mental-health VSP deployments where TTS/ASR and turn-taking issues can erode realism and rapport [50]. Such frictions also fit broader VR findings: presence can raise involvement yet increase workload when pacing and user control are not well managed [30], [31].

B. Positioning AI-VSP as a Supplemental Preparation Modality

In response to RQ2, no statistically significant differences emerged between the SP encounter performance scores of AI-VSP users and traditionally prepared peers, supporting performance parity rather than replacement superiority. In transfer-of-training terms, the AI-VSP provides sufficient task fidelity and cue overlap to support near transfer to the SP encounter, but—absent extended practice and multi-case variability—should be positioned as an adjunct that strengthens readiness rather than a standalone substitute [59], [60]. Deliberate practice similarly argues for repeated, goal-directed rehearsal with immediate feedback and progressive challenge; a single session can catalyze improvement, yet durable gains typically require distributed, feedback-rich cycles that complement—not replace—high-stakes SP encounters [61], [62].

This interpretation aligns with the SP/VSP literature: despite their costs and logistical burdens, SPs remain the preferred modality for nuanced affect, adaptive feedback, and assessment validity. [7], [8], [40]. VSP or AI-VSP have demonstrated value for history-taking and communication practice, particularly for access and standardization, while noting constraints around conversational fidelity and usability [42], [45], [48], [50]. Meta-analyses and reviews of immersive learning in nursing and higher education further recommend explicit design elements—authentic tasks, pacing, and feedback—and often endorse

blended models to maximize effectiveness and scalability [33], [34], [35], [36]. Early AI-VSP reports echo these themes: learners perceive realism and utility, but outcomes hinge on dialogue control, speech quality, and latency [13], [14], [15].

C. Post-AI-VSP Encounter Gains in Self-Efficacy

In relation to RQ3, learners showed a gain in self-efficacy—from 3.74 to 4.11—after a single AI-VSP encounter session exposure ($p < 0.001$), signifying a meaningful short-term effect on a validated scale [68]. Such short-term confidence improvements are consistent with reports that immersive technologies in nursing and higher education can enhance affective outcomes alongside skills when key design elements are explicit [33], [34], [35]. From a transfer-of-training perspective, rising self-efficacy signals that practice tasks provided enough fidelity and cue overlap to strengthen expectations of successful performance in the SP encounter setting [59], [60].

Qualitative accounts illuminate mechanisms behind this shift: learners cited natural, open-ended dialogue, persona diversity, and structured assistances as mechanisms that helped them organize questions, redirect digressions, and confirm information—microconditions aligned with deliberate practice, where clear goals, focused repetition, and informative feedback drive improvement [61], [62]. Early AI-VSP studies similarly note perceived realism and utility when conversational control and speech quality are adequate [13], [14], [55].

D. Dissociation Between Experience and AI-VSP Encounter Performance

Despite moderate–high ratings of engagement, presence, and immersion—and strong intercorrelations among these constructs ($r = 0.63, 0.56, 0.83$)—their associations with immediate AI-VSP encounter performance were near zero ($r = -0.17, 0.07, -0.11$). This pattern aligns with prior VR research showing that presence can heighten affect yet also elevate workload when pacing and user control are limited, which can depress task coverage [27], [29], [30], [31]. It also echoes reports from VSP platform where conversational mechanics—latency, over-long turns, limited interruptibility, and synthetic prosody—erode realism and efficiency even when users feel “immersed” [48], [49], [50], and with recent AI-VSP findings on the importance of dialogue control for perceived fidelity [13].

Linking to RQ4, group-level parity on subsequent SP encounter performance suggests that beneficial transfer can occur even when subjective experience does not track immediate AI-VSP encounter performance. From a transfer-of-training view, high presence without faithful task demands and controllability may not strengthen cue–response mappings needed for near transfer [59], [60]. Deliberate practice further implies that gains depend on tightly paced, feedback-rich repetitions toward clear goals; affective absorption alone is insufficient if learners cannot steer turn-taking to test and refine questioning strategies [61], [62].

E. Mechanisms: Facilitators and Barriers to Learning

In relation to RQ5, interviews clarified how the learning platform supported—or impeded—learning. Four facilitators emerged.

- Concise, interruptible dialogue restored learner agency during turn-taking, enabling targeted question–answer cycles central to deliberate practice and to building cue–response mappings for near transfer [41], [59], [61].
- Low ASR and TTS latency with more natural prosody improved conversational fidelity and rapport, consistent with guidance from VSP and AI-VSP research [13], [48], [49].
- Persona fit and case variability required adaptive questioning across communicative styles, aligning with transfer principles that emphasize varied practice around invariant task cues [59], [60].
- Structured assistance aided orientation without over-structuring, echoing simulation and higher-ed VR design guidance on authenticity, pacing, and minimal interface overhead [5], [33].

Barriers mapped closely to the inverse of these levers. Over-verbosity, weak interruptibility, synthetic prosody, and recognition errors inflated workload and eroded perceived realism—well-documented threats in VSP platforms and immersive mental-health encounters [48], [49], [50]. Visual freezes and minor headset frictions further disrupted pacing and attention, a known risk when presence is high but control is limited [30], [31], [37]. Together, these frictions help explain why experience ratings were high yet uncoupled from immediate AI-VSP encounter performance (RQ4) and why the AI-VSP functioned best as a supplement in the preparation sequence (RQ2).

F. Implications for AI-for-VR Design and Curriculum

Taken together, these findings suggest two practical directions—system design and curriculum integration—that can make AI-for-VR training both usable and educationally effective. Through our analysis, we make the following recommendations:

System design:

- Cap agent replies to about one to three sentences, provide reliable interruption (push-to-talk and barge-in), and minimize ASR/TTS delay while improving prosody so learners can steer turn-taking and maintain conversational fidelity [13].
- Keep the clinic single-user with straightforward controls and add structured assistance—live captions and concise cue sheets tied to the case rubric—to focus attention without scripting [33].
- Organize practice as short, repeatable bouts with rapid restarts, calibrated persona difficulty, and concise, rubric-aligned feedback to support deliberate practice and near transfer [59], [61].
- Track task-aligned indicators (e.g., rubric coverage, questioning efficiency, error recovery) alongside usability; avoid treating presence/immersion as performance proxies; use SUS for usability and maintain inter-rater reliability for scoring [30], [66].

- Embed governance: disclose AI use, protect privacy with de-identified logs, monitor bias, and provide faculty authoring workflows for updating cases and personas [53].

Curriculum integration:

- Use AI-VSP encounter sessions as low-stakes, on-demand warm-ups before SP encounters and guided debriefs to blend scalability with the pedagogical strengths of SP encounter programs [8].
- Distribute several micro-sessions (short sessions that reduce cognitive load) across a module, vary cases and persona difficulty to support generalization, and pair practice with brief reflection or automated, rubric-aligned feedback [61].
- Align assessments with training by using the same rubric in AI-VSP and SP encounter settings, report both experience and performance outcomes, and offer faculty development on case authoring, calibration, and responsive use [33].

G. Limitations and Future Work.

There were limitations in this study. First, the single-institutional and self-selected design with a nonequivalent (in number) control may have introduced selection bias that could limit external validity. Second, the AI-VSP exposure was brief in nature. Hence, the results may not generalize to longer training scenarios in other curricula. Third, the technical constraints—ASR/TTS latency, synthetic prosody, occasional freezes, and headset frictions—may have reduced conversational fidelity and rapport.

Future work should strive to strengthen external validity by using multi-site recruitment and randomized designs with more balanced group sizes. Longer and repeated AI-VSP encounter practice with follow-up measures may benefit determining whether constructs of interest such as self-efficacy, engagement, and others, as well as knowledge transfer will persist or change over time and across curricula. Finally, it is anticipated that, with the advancement of technologies, the obstacles of technical limitations—such as ASR/TTS latency, synthetic prosody, occasional freezes, and headset comfort—can be tackled to improve conversational fidelity and rapport.

VI. CONCLUSION

This study introduces an AI-VSP integrated into an immersive clinic and demonstrates that it is a usable, confidence-building supplement for patient encounter training. Usability was excellent ($SUS = 81.6/100$), and a single session increased self-efficacy (3.74 to 4.11 , $p < 0.001$). SP encounter performance scores for AI-VSP users were comparable to traditionally prepared peers, supporting a supplemental—rather than replacement—role. Notably, engagement, presence, and immersion inter-correlated but did not track immediate AI-VSP encounter performance, underscoring that controllability and dialogue mechanics matter more than added “immersion” for task coverage.

Qualitative evidence clarified the mechanisms: learning was facilitated by concise, interruptible replies, low speech-pipeline

latency with natural prosody, appropriate persona fit, and structured assistance; it was hindered by verbosity, weak turn-taking control, synthetic voices, misrecognitions, and headset frictions.

These findings translate into concrete design and curricular guidance: cap reply length, enable reliable barge-in, tighten ASR/TTS timing and prosody, retain simple single-user controls, and pair short AI-VSP encounter sessions with brief, rubric-aligned feedback and varied cases. In sum, AI-VSPs offer a scalable pathway to deliberate, on-demand rehearsal that complements SP encounters and supports transfer-ready preparation for nursing communication.

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